Experimental Methods in Fluid Mechanics
Particle Image Velocimetry
M2 - IPParis

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Go to : `https://147.250.144.XXX` using the code you have been given
In `Application/Others/`, you will find a Matlab instance embedding PIVLab application. Open it.

1 Discovering the tool

PIVLab has been designed to provide all the relevant features of a PIV software, from acquisition to basic statistical analysis. In this session, we will focus on PIV processing from existing images. The statistical analysis will be done with Matlab.

Importing images : use the Import button to import images. Choose any datasets.

Perform PIV analysis : this is done in two steps.

- First set the PIV settings either from the Analysis menu, either from the parameter wheel in the quick access tool bar.
- Choose any parameters and start the analysis.
- Provide comments on your results and share them with the class.

Calibration : open the calibration tool, either from the Calibration menu, either from the parameter wheel in the quick access tool bar.

- Play with the different options and parameters.
- See the effects on the computed fields.

Validation : open the validation tool from the Validation menu. Choose the velocity based validation.

- Play with the different options and parameters.
- See the effects on the computed fields.
- Provide comments on your results and share them with the class.

Basic plots : there are several ways of plotting data with PIVLab.

- Browse the Plot menu and try different options
- Browse the Extraction menu and try different options
- Browse the Statistics menu and try different options
- Share your findings with the class

2 Choosing PIV parameters

The two main features are single and multi-pass. We have explained the differences, but here, we will play with them.

Loading datasets : we will import and browse a specific datasets.

— From the file menu, start a new session
— Import the Fuerteventura datasets
— Browse the images and describe the expected flow

Single pass : this the basic PIV method. Let's explore it first.

— Use increasing windows sizes to analyze the data (from 8 to 128 pixels). Do not use overlap for the smallest windows.
— Carefully look at the computed velocity fields and write down your observations.
— Compute validation rates.
— Comment on the results : what are the differences between the different windows sizes ? Why is it always so poor ? Are the validation rates always in line with your observations ?

Multi-pass : the best way to turn the problem raised above is to perform multi-pass analysis.

— From the observations you did at the previous question, choose relevant parameters for the multi-pass analysis.
— Look at the computed velocity fields
— Compute validation rates
— Comment on the results

3 More advanced plotting with Matlab

Start a new session, choose a data-set that suits you and perform a PIV analysis with relevant parameters. Open the Matlab script `piv_lab_TP.m` to get some ideas. Feel free to invent and to share your findings with others.

Exporting to Matlab workspace : in the file menu, use the Export option "All results to Matlab workspace". Alternatively, you can also export as a Matlab file for further loading. You get several velocity variables :

— `u_original`, `v_original` : the original velocity fields, output of your PIV process
— `u_filtered`, `v_filtered` : the filtered velocity fields, output of your validation process
— `u_smoothed`, `v_smoothed` : the smoothed velocity fields, output of your smoothing process (`Plot\Spatial`)
— `u_component`, `u_component` :??
— `typevector_filtered` : 0 for masked, 1 for regular and 2 for filtered vectors
— various other self explanatory fields that are filled if you have calculated them

Plotting movies : the given script `piv_lab_TP.m` proposes several ways of plotting movies. Play with that and share your findings with the others.

Filtering relevance : the filtering plays a crucial role in PIV, monitoring its level is crucial to have a good idea of its impact on the data but also to assess the whole acquisition chain quality.

— find a way to plot the evolution of the number of filtered data points in your results
— use this for several PIV settings and draw your conclusions